Package ‘modeldiag’

June 2, 2026

Type Package

Title Comprehensive Diagnostics for Statistical Models

Version 0.1.2

Date 2026-06-02

Description

Provides a unified framework for diagnosing common issues in statistical models including linear models, generalized linear models (logistic and Poisson regression), and survival models. Implements tests for multicollinearity, heteroscedasticity, autocorrelation, normality, influential observations, overdispersion, zero-inflation, and proportional hazards assumptions. Includes visualization methods for graphical diagnostics. Methods are based on established approaches including Fox and Monette (1992) <doi:10.1080/01621459.1992.10475190>, Breusch and Pagan (1979) <doi:10.2307/1911963>, and Dean and Lawless (1989) <doi:10.1080/01621459.1989.10478792>.

Maintainer Emmanuel Adewuyi <emmanuel.adewuyi@lshtm.ac.uk>

License MIT + file LICENSE

Encoding UTF-8

URL https://github.com/Teniola17/modeldiag,

https://teniola17.github.io/modeldiag/

BugReports https://github.com/Teniola17/modeldiag/issues

Depends R (>= 3.5.0)

Imports stats, graphics, car, lmtest, ResourceSelection, survival

Suggests testthat (>= 3.0.0), knitr, rmarkdown

RoxygenNote 7.3.3

VignetteBuilder knitr

Config/testthat/edition 3

NeedsCompilation no

Author Emmanuel Adewuyi [aut, cre] (ORCID:
<https://orcid.org/0000-0002-7920-6866>),
Adewale Lukman [aut] (ORCID: <https://orcid.org/0000-0003-2881-1297>),
Abiola Owolabi [aut] (ORCID: <https://orcid.org/0000-0001-9843-5085>)

Contents

`modeldiag-package` *modeldiag: Comprehensive Diagnostics for Statistical Models*

Description

Provides a unified framework for diagnosing common issues in statistical models including linear models, generalized linear models (logistic and Poisson regression), and survival models. Implements tests for multicollinearity, heteroscedasticity, autocorrelation, normality, influential observations, overdispersion, zero-inflation, and proportional hazards assumptions. Includes visualization methods for graphical diagnostics. Methods are based on established approaches including Fox and Monette (1992) doi:10.1080/01621459.1992.10475190, Breusch and Pagan (1979) doi:10.2307/1911963, and Dean and Lawless (1989) doi:10.1080/01621459.1989.10478792.

Author(s)

Maintainer: Emmanuel Adewuyi <emmanuel.adewuyi@lshtm.ac.uk> (ORCID)

Authors:

- Adewale Lukman <adewale.lukman@und.edu> (ORCID)
- Abiola Owolabi <abiola.owolabi@lautech.edu.ng> (ORCID)

See Also

Useful links:

- https://github.com/Teniola17/modeldiag
- https://teniola17.github.io/modeldiag/
- Report bugs at https://github.com/Teniola17/modeldiag/issues

---

check_autocorrelation *Check Autocorrelation*

Description

Performs the Durbin-Watson test for first-order autocorrelation in residuals.

Usage

```
check_autocorrelation(model)
```

Arguments

`model`	A fitted lm object.

Value

An htest object or NA if computation fails.

Examples

```
model <- lm(mpg ~ wt + hp, data = mtcars)
check_autocorrelation(model)
```

---

check_box_tidwell *Check Logistic Linearity with Box-Tidwell Test*

Description

Performs a Box-Tidwell test for linearity of continuous predictors in the logit.

Usage

```
check_box_tidwell(model)
```

Arguments

`model`	A fitted glm object.

Value

A Box-Tidwell test result or NA if computation fails.

Examples

```
model <- glm(am ~ wt + hp, data = mtcars, family = binomial)
check_box_tidwell(model)
```

`check_functional_form_coxph` *Check Cox Model Functional Form*

Description

Provides guidance for checking nonlinear functional form in Cox models using martingale residuals.

Usage

```
check_functional_form_coxph(model)
```

Arguments

`model`	A fitted coxph object.

Value

A character message or NA if computation fails.

Examples

```
if (requireNamespace("survival", quietly = TRUE)) {
  model <- survival::coxph(
    survival::Surv(time, status) ~ age + sex,
    data = survival::lung
  )
  check_functional_form_coxph(model)
}
```

`check_heteroskedasticity` *Check Heteroskedasticity*

Description

Performs Breusch-Pagan test for heteroskedasticity.

Usage

```
check_heteroskedasticity(model)
```

Arguments

`model`	A fitted lm object.

Value

An htest object or NA if computation fails.

Examples

```
model <- lm(mpg ~ wt + hp, data = mtcars)
check_heteroskedasticity(model)
```

---

check_hosmer_lemeshow *Check Logistic Goodness of Fit*

Description

Performs the Hosmer-Lemeshow goodness-of-fit test for binomial glm models.

Usage

```
check_hosmer_lemeshow(model)
```

Arguments

model	A fitted binomial glm object.

Value

An htest object or NA if computation fails or the model is not binomial.

Examples

```
model <- glm(am ~ wt + hp, data = mtcars, family = binomial)
check_hosmer_lemeshow(model)
```

---

check_influential_coxph *Check Influential Observations in Cox Models*

Description

Identifies influential observations in Cox models using dfbeta residuals.

Usage

```
check_influential_coxph(model)
```

Arguments

model	A fitted coxph object.

Value

A list containing dfbeta residuals and influential point indices, or NA if computation fails.

Examples

```
if (requireNamespace("survival", quietly = TRUE)) {
  model <- survival::coxph(
    survival::Surv(time, status) ~ age + sex,
    data = survival::lung
  )
  check_influential_coxph(model)
}
```

---

check_influential_glm *Check Influential Observations in GLMs*

Description

Identifies influential observations in generalized linear models using Cook's distance.

Usage

```
check_influential_glm(model)
```

Arguments

model	A fitted glm object.

Value

A list containing Cook's distances and influential point indices.

Examples

```
model <- glm(am ~ wt + hp, data = mtcars, family = binomial)
check_influential_glm(model)
```

---

check_linearity *Check Linearity*

Description

Performs Ramsey's RESET test for linear model misspecification.

Usage

```
check_linearity(model, power = 2)
```

Arguments

model	A fitted lm object.
power	Powers of fitted values to include in the test.

Value

An htest object with an added note, or NA if computation fails.

Examples

```
model <- lm(mpg ~ wt + hp, data = mtcars)
check_linearity(model)
```

---

check_normality *Check Normality of Residuals*

Description

Performs the Shapiro-Wilk test on model residuals.

Usage

```
check_normality(model)
```

Arguments

`model`	A fitted model object.

Value

An htest object or NA if computation fails.

Examples

```
model <- lm(mpg ~ wt + hp, data = mtcars)
check_normality(model)
```

---

check_outliers *Check Influential Observations*

Description

Identifies influential observations using Cook's distance.

Usage

```
check_outliers(model, cutoff = 4/length(residuals(model)))
```

Arguments

`model`	A fitted model object.
`cutoff`	Cook's distance cutoff. Defaults to 4 divided by the number of observations.

Value

A list containing Cook's distances and influential point indices.

Examples

```
model <- lm(mpg ~ wt + hp, data = mtcars)
check_outliers(model)
```

---

check_overdispersion *Check Overdispersion*

Description

Checks whether a Poisson glm appears overdispersed using the residual deviance to degrees-of-freedom ratio.

Usage

```
check_overdispersion(model)
```

Arguments

`model`	A fitted Poisson glm object.

Value

A list with overdispersion diagnostics, or NA if the model is not Poisson or computation fails.

Examples

```
model <- glm(carb ~ wt + hp, data = mtcars, family = poisson)
check_overdispersion(model)
```

---

check_proportional_hazards *Check Proportional Hazards*

Description

Performs a proportional hazards test for a fitted Cox model using Schoenfeld residuals.

Usage

```
check_proportional_hazards(model)
```

Arguments

`model`	A fitted coxph object.

Value

A cox.zph result or NA if computation fails.

Examples

```
if (requireNamespace("survival", quietly = TRUE)) {
  model <- survival::coxph(
    survival::Surv(time, status) ~ age + sex,
    data = survival::lung
  )
  check_proportional_hazards(model)
}
```

`check_residual_analysis` *Check Residual Summary*

Description

Summarizes deviance residuals for a generalized linear model.

Usage

```
check_residual_analysis(model)
```

Arguments

`model`	A fitted model object.

Value

A list of residual summary statistics, or NA if computation fails.

Examples

```
model <- glm(carb ~ wt + hp, data = mtcars, family = poisson)
check_residual_analysis(model)
```

`check_separation` *Check Separation in GLMs*

Description

Checks for simple indicators of complete or quasi-complete separation.

Usage

```
check_separation(model)
```

Arguments

`model`	A fitted glm object.

Value

A character message describing whether separation was detected, or NA if the model is not a glm.

Examples

```
model <- glm(am ~ wt + hp, data = mtcars, family = binomial)
check_separation(model)
```

---

check_vif   *Check Variance Inflation Factors*

Description

Computes variance inflation factors to detect multicollinearity.

Usage

```
check_vif(model)
```

Arguments

`model`	A fitted model object.

Value

A list describing whether VIF computation succeeded. Successful results include a ‘vif‘ element containing the variance inflation factors.

Examples

```
model <- lm(mpg ~ wt + hp + disp, data = mtcars)
check_vif(model)
```

---

check_zero_inflation   *Check Zero Inflation*

Description

Compares observed zeros with the number expected under a fitted Poisson glm.

Usage

```
check_zero_inflation(model)
```

Arguments

`model`	A fitted Poisson glm object.

Value

A list with zero-inflation diagnostics, or NA if the model is not Poisson or computation fails.

Examples

```
model <- glm(carb ~ wt + hp, data = mtcars, family = poisson)
check_zero_inflation(model)
```

---

diagnose_model.glm *Diagnose Statistical Models*

Description

This is a generic function for performing diagnostic checks on statistical models. It dispatches to specific methods based on the model type.

Usage

```
## S3 method for class 'glm'
diagnose_model(model, ...)

## S3 method for class 'lm'
diagnose_model(model, ...)

## S3 method for class 'coxph'
diagnose_model(model, ...)

diagnose_model(model, ...)
```

Arguments

model	A fitted model object.
...	Additional arguments passed to specific methods.

Value

An object of class "model_diagnostics" containing the results of various diagnostic tests.

Examples

```
# Linear model diagnostics
model_lm <- lm(mpg ~ wt + hp, data = mtcars)
diag_lm <- diagnose_model(model_lm)
summary(diag_lm)
plot(diag_lm)

# Logistic regression diagnostics
model_glm <- glm(am ~ wt + hp, data = mtcars, family = binomial)
diag_glm <- diagnose_model(model_glm)
summary(diag_glm)

# Poisson regression diagnostics
model_pois <- glm(carb ~ wt + hp, data = mtcars, family = poisson)
diag_pois <- diagnose_model(model_pois)
summary(diag_pois)
```

```
# Cox proportional hazards diagnostics
library(survival)
data(lung)
model_cox <- coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
diag_cox <- diagnose_model(model_cox)
summary(diag_cox)
```

`plot.model_diagnostics` *Plot Model Diagnostics*

Description

Generates diagnostic plots for the fitted model.

Usage

```
## S3 method for class 'model_diagnostics'
plot(x, ...)
```

Arguments

x	An object of class "model_diagnostics".
...	Additional arguments passed to plotting functions.

Value

None (plots are displayed).

`print.model_diagnostics` *Print Model Diagnostics*

Description

Prints a summary of the model diagnostics object.

Usage

```
## S3 method for class 'model_diagnostics'
print(x, ...)
```

Arguments

x	An object of class "model_diagnostics".
...	Additional arguments passed to print.

Value

The object x, invisibly.

`summary.model_diagnostics`

Summarize Model Diagnostics

Description

Provides a detailed summary of diagnostic test results.

Usage

```
## S3 method for class 'model_diagnostics'
summary(object, ...)
```

Arguments

`object`	An object of class "model_diagnostics".
`...`	Additional arguments (currently ignored).

Value

The object, invisibly.

Index